A Comparative Evaluation of Extractive and Abstractive Medical Summarization Methods

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1 Research Question

How does the choice between extractive and abstractive summarization affect factual reliability, ease of understanding and overall summary quality in medical text summarization?

2 Database

https://huggingface.co/datasets/BioLaySumm/BioLaySumm2025-PLOS
https://huggingface.co/datasets/BioLaySumm/BioLaySumm2025-eLife

3 Reference Papers

https://pmc.ncbi.nlm.nih.gov/articles/PMC13142433/
https://pmc.ncbi.nlm.nih.gov/articles/PMC10449915/

4 Methodology

Generate summaries from the articles using:

- TF-IDF
- TextRank
- pre-trained Hugging Face model
  https://huggingface.co/Falconsai/medical_summarization
  https://huggingface.co/Mahalingam/med-summary
  https://huggingface.co/AbhinavSK/cikista-medical-report-summarizer
- LLM
  https://github.com/yashaswi1306/Flashcard-Generator-RAG
  model used=Qwen/Qwen2.5-3B-Instruct

Evaluate results based on:

- ROUGE-1, ROUGE-2, ROUGE-L
- Python rule-based checker for keyword coverage and regex matching for numeric values
- Human evaluation